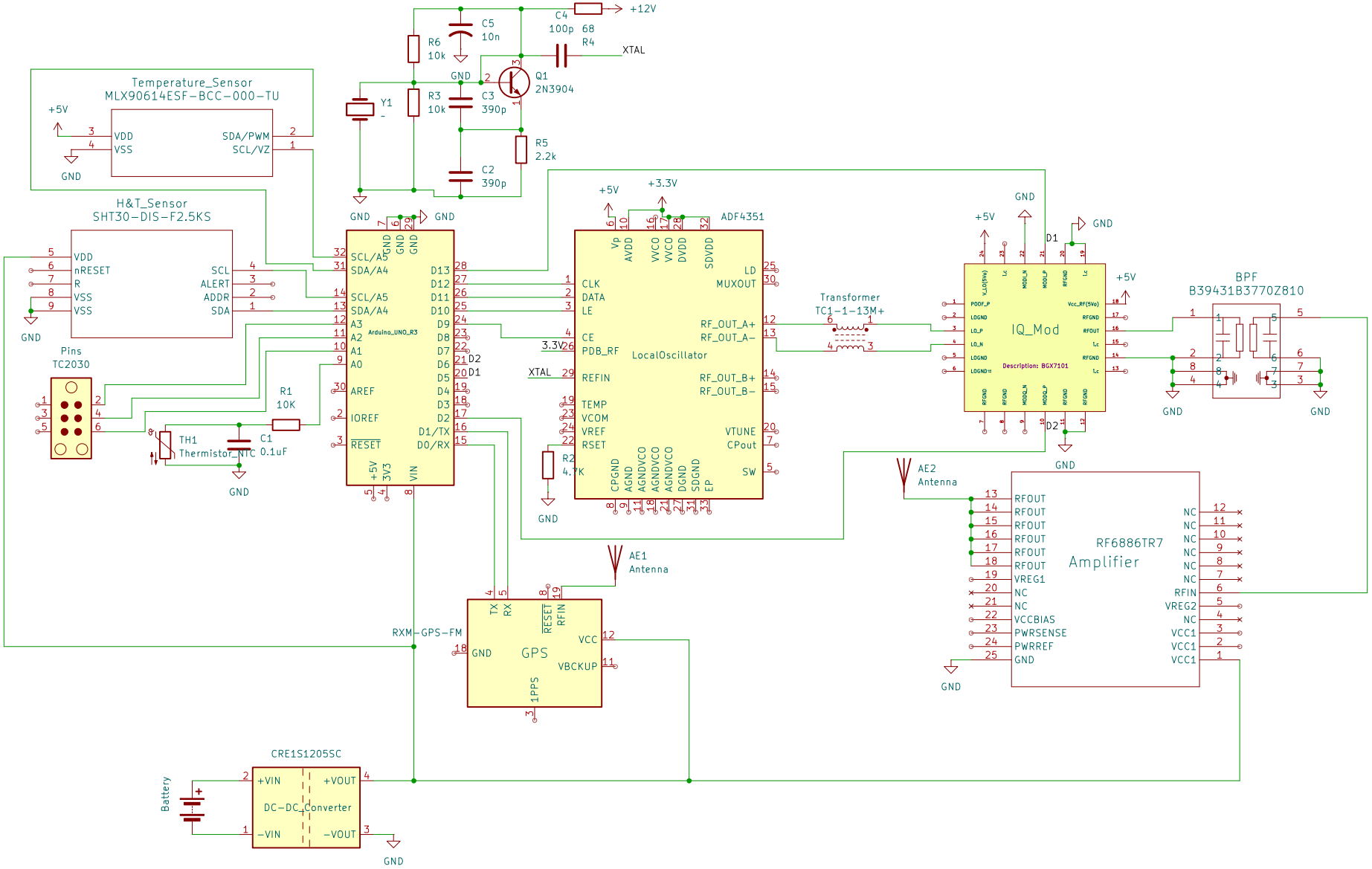

Temperature_Sensor
MLX90614ESF-BCC-000-TU
+5V
VDD
VSS
SDA/PWM
SCL/VZ
GND
H&T_Sensor
SHT30-DIS-F2.5KS
VDD
nRESET
R
VSS
VSS
SCL
ALERT
ADDR
SDA
GND
Pins
TC2030
TH1
Thermistor_NTC
R1
10K
C1
0.1uF
GND
Arduino_UNO_R3
SCL/A5
SDA/A4
A3
A2
A1
A0
AREF
IOREF
RESET
+5V
3V3
VIN
D13
D12
D11
D10
D9
D8
D7
D6
D5
D4
D3
D2
D1/TX
D0/RX
D2
D1
GND
GND
C5
10n
R6
10k
R3
10k
C3
390p
C2
390p
Y1
~
GND
Q1
2N3904
C4
100p
68
R4
XTAL
+12V
R5
2.2k
+5V
+3.3V
ADF4351
Vp
AVDD
VVCO
VVCO
DVDD
SDVDD
CLK
DATA
LE
CE
PDB_RF
REFIN
TEMP
VCOM
VREF
RSET
LD
MUXOUT
RF_OUT_A+
RF_OUT_A-
RF_OUT_B+
RF_OUT_B-
VTUNE
CPout
SW
LocalOscillator
3.3V
XTAL
R2
4.7K
CPGND
AGND
AGNDVCO
AGNDVCO
AGNDVCO
DGND
SDGND
EP
GND
Transformer
TC1-1-13M+
+5V
GND
D1
IQ_Mod
Description: BGX7101
D2
GND
BPF
B39431B3770Z810
GND
GND
AE2
Antenna
GND
RF6886TR7
Amplifier
RFOUT
VREG1
NC
VCCBIAS
PWRSENSE
PWRREF
GND
RFIN
VREG2
VCC1
AE1
Antenna
RXM-GPS-FM
GPS
TX
RX
RESET
RFIN
GND
VCC
VBCKUP
1PPS
CRE1S1205SC
Battery
+VIN
-VIN
+VOUT
-VOUT
DC-DC_Converter
GND

1.Humidity & Temperature Sensor

SHT30-DIS:

The **SHT30-DIS** is a high-accuracy, digital sensor used for measuring relative humidity and ambient temperature.

Alternatives: HS3001, SHT21P, SHT31-ARP, etc.

2.Open Temperature Sensor

MLX90614: Infrared (Non-Contact) Temperature Sensor:

The **MLX90614** (U6) is an "open" high-range temperature sensor that measures surface temperature without making physical contact.

- **How it Works:** It detects infrared energy emitted by an object to calculate its temperature.
- **Communication:** Like the SHT30, it operates over the **I2C bus**.
- **Purpose:** Used to monitor external targets or extreme heat sources.

Alternatives: MCP9700, TMP235, MAX 31826,etc.

3. DC-DC Converter

CRE1S1205SC (12V to 5V):

The CRE1S1205SC is a 1-Watt, isolated DC-DC converter designed to provide a stable, regulated output from a potentially fluctuating battery source.

Alternatives: LM2596, MP1584, LTC2945 ,etc.

4. IQ Modulator

BGX7101:

It takes your baseband "Data" signals (I and Q) and the "Carrier Frequency" (LO) from the synthesizer and mixes them together to create the modulated RF signal at 433 MHz.

- **LO Splitting:** It takes the Local Oscillator (LO) signal and splits it into two paths: an "In-phase" and a "Quadrature" 90 ° path.

- **Dual Mixing:** The **I (In-phase)** data is mixed with the 0° carrier, and the **Q (Quadrature)** data is mixed with the 90°carrier.
- **Summation:** These two signals are then summed together to produce a single RF output that contains all your data, ready for transmission.

Alternatives: ADL5375, TRF370417

5. Band Pass Filter

B39431B3710U410 (B3710):

It acts as a "Gatekeeper" for frequency. It allows **only** signals around **433.92 MHz** to pass through and blocks everything else (noise, harmonics).

Alternatives: Tai-Saw Technology (TST) TA0230A, Murata SAFB433MAA0F0A

6. High Power Linear Amplifier

RF6886TR7:

It takes the weak signal from the modulator/filter and boosts it to high power (up to 3.6 Watts) for transmission.

Operating Frequency: 100MHz to 1000MHz (covers 433MHz target)

Alternatives: Qorvo QPA9510, Skyworks SKY65162-70LF

7. RF Transformer:

TC1-1-13M+:

It acts as a **Coupling Transformer**. It takes the differential Local Oscillator (LO) signal from the Synthesizer and passes it to the Modulator while providing **DC Isolation**. This prevents the DC bias voltage of one chip from damaging or interfering with the other, while allowing the high-frequency RF signal to pass through.

Alternatives: TC1-1+, MACOM MABA-007159-000000, Johanson Technology 0433BM15A0001

8.XTAL Circuit:

The XTAL circuit in the image is a discrete oscillator designed to provide a high-precision, stable clock signal. The output of this circuit is fed into the REFIN (Pin 29) of ADF4351 Local Oscillator to serve as its frequency reference.

Alternatives: XO, TCXO ,VCXO ,OCXO ,etc.

9. Local Oscillator

ADF4351:

This chip generates the radio frequency "carrier" wave. It outputs the clean sine wave (programmable via software) that travels through the transformer to the modulator to be mixed with data.

- **AC Coupling:** Blocks the DC Signal.
- **Input Impedance:**The discrete oscillator must be designed to drive this capacitive load without Frequency Pulling.

Alternatives: Maxim (Analog Devices) MAX2870, Texas Instruments LMX2572

10. GPS

Linx RXM-GPS-FM:

This is a high-sensitivity, surface-mount **GPS Receiver Module**. It is designed to track satellites and provide precise position (Latitude, Longitude, Altitude) and time data to your microcontroller.

Alternatives: u-blox NEO-M8N, Quectel L80 / L86

11. Arduino UNO

The **Arduino Uno R3** is the main digital controller in the project. It acts as the "brain" that orchestrates the entire telemetry system.

- **Processor:** Microchip (Atmel) **ATmega328P** (8-bit AVR).
- **Clock Speed:** 16 MHz.
- **Memory:** 32 KB Flash (Program), 2 KB SRAM (Variables), 1 KB EEPROM.
- **Voltage:** 5V Logic (This is critical because **BGX7101** and **ADF4351** control lines interact with this voltage).
- **Role in Schematic:**

- **I2C Master:** It reads temperature/sensor data from **U2** (SHT30) and **U6** (MLX90614) via the SDA/SCL lines.
- **SPI Master:** It configures the frequency of **U5** (ADF4351 Synthesizer) via the CLK/DATA/LE lines.
- **Data Generator:** It generates the baseband "Data" signals (D0-D13) that are sent to the IQ Modulator.